

Lean Transformation Framework for Software Intensive Companies

Responding to Challenges Created by the Cloud

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Abstract—Today, the cloud phenomenon is challenging companies' business and revenue models. Many enterprises are starting a fundamental change in order to benefit from the opportunities and to respond to the challenges that the cloud brings. At the same time, interest in lean thinking has grown in industries and research communities. Lean paradigm is assessed as a potential means to help companies pursue efficiency. This paper builds on premise that in order to succeed, software intensive enterprise's cloud transformation should have lean thinking woven in – we talk about lean transformation. This paper extends the current available lean related literature that, however, is mainly intended to manufacturing industries. This paper proposes a lean transformation framework, which highlights the significance of learning, iterative execution and holistic approach involving the whole enterprise in the transformation. First experiences of using the proposed lean transformation framework in a large IT service company are described as well.

Keywords—Lean Paradigm; Transformation; Cloud Phenomenon

I. INTRODUCTION

“Cloud” and “lean” are contemporary paradigms. Prevailing definitions of cloud / cloud computing have been studied, but conclusion is that clouds do not have a clear and complete definition [1]. However, a proposal for the definition of clouds is: clouds are a large pool of easily usable and accessible resources, which can be dynamically re-configured [1]. In this paper, cloud phenomenon is thought to have an impact also on other areas of businesses and brings changes to the external business environment. These changes will present both opportunities and threats, with some possessing the potential to radically alter how an Information and Communication Technology (ICT) company conducts its business. In general, “cloud forces” are beyond the direct control of an organisation. Instead, companies must recognise the cloud as something to be prepared for and responded to.

Utilising opportunities of the cloud is especially useful when composing new services. Service science is the study of service systems (e.g. [2]), which comprise people and technologies that adaptively compute and adjust to a system's

changing value of knowledge [3]. Service science combines organisation and human understanding with business and technological understanding in order to explain the origins and growth of service systems, to solve fundamental problems, and to produce unique service professionals and service scientists [2]. Service science is grounded in the cross-functional issues of business and its operating processes, and its theoretical roots extend to engineering, technology, and the social sciences [4].

Goals towards efficiency, cost-effectiveness, good quality and superior user experience are on the agenda of contemporary ICT firms. Since the mid 1990s, agile methods have been in use in software intensive companies to achieve these goals (e.g. [5], [6]). However, agile methods are not sufficient, and moreover, the cloud brings along new challenges. Lean thinking (e.g. [7]) that has been introduced in several industries, also in software development [8], [9], is anticipated to help companies in the challenges described above.

One of the lean principles highlights the importance of optimising the whole (e.g. [9]) and when adopted in enterprise level, the sought-for result is lean enterprise. According to several authors (see e.g. [10], [11]) evolution of an enterprise towards a lean enterprise cannot be managed with a top-down development strategy, but also a bottom-up approach is needed. Despite the existing frameworks and guidelines for organisational lean transformation (e.g. [7], [10], [12]), further studies are needed to revise the frameworks to better apply in the contemporary companies – especially in software intensive companies.

The research questions of this paper are:

- What steps an organisation needs to take in order to transform to lean?
- What viewpoints shall be considered in the transformation process?

The purpose of this paper is to propose a lean transformation framework for software intensive companies, who aim to utilise the opportunities created by the cloud. The proposed framework is based on the existing models ([7], [10], and [13]) that are intended mainly to manufacturing industries,

though. First experiences on using the proposed lean transformation framework are presented as well. This study points out the importance of a transformation framework, which has a holistic view and works as a roadmap for the change in question. This study utilises definitions and views of the lean paradigm (e.g. [7], [8], and [9]).

This study follows the paradigm of the design science [14], which pursues knowledge and understanding of a problem domain and its solution by building and applying of the designed artefact. The evaluation of the artefact provides feedback information and a better understanding of the problem in order to improve both the artefact and the design process. In this research the artefact is the transformation framework.

Section II of this paper presents the background for this study. In section III the proposed framework is presented and explained in details. Experiences of using the framework are presented in section IV. The last two sections conclude the paper with some observations on the limitations of the study and an outline of avenues for future research.

II. THEORETICAL BASIS FOR TRANSFORMATION FRAMEWORKS

A. Setting the Scene

Enterprise transformation concerns a fundamental change that can involve new or old value propositions in terms of products and services, how they are delivered and supported, and how the organisation is built to provide these offerings. Transformation is driven by value deficiencies and involves examining and changing work processes. Transformation needs both strategy-oriented (i.e. initiatives driven by external factors) and operations-oriented (i.e. initiatives driven by internal factors) approaches. [15].

Conclusions from case studies of the ways how transformation has been implemented are described ([15], [16]) by perspectives of *goal*, *approach*, and the *scope*. The *goal or ends* of transformation can range from cost efficiencies to fundamental changes in markets, *the approach or means* of transformation can range from upgrading people's skills to major changes in strategy, and *the scope* of transformation can range from work activities to the enterprise as a whole. The ends and means influence the scope of transformation. The costs and risks of transformation increase when moving towards wider scope [16].

B. Lean Transformation

Lean thinking stems from automotive industry, and later it has been adopted in other manufacturing industries and further to lean enterprise [7] more generally. Within the last decade, software development challenges and the applicability of lean principles to software development have been studied [17]. The five lean principles, i.e. identify value, map the value stream, create flow, establish pull, and seek perfection, were applied to software development [17]. It is argued that the software industry is the most suited of all the industries to transform to the lean production [17], and e.g. Poppendieck & Poppendieck [8], [9] concluded with the following seven principles of lean

software development: 1) eliminate waste, 2) build quality in, 3) create knowledge, 4) defer commitment, 5) fast delivery, 6) respect people, and 7) optimise the whole.

The lean principles listed above are more as a philosophy than practical guidelines. Applying them in an organisation most evidently needs a big change. Thus, if a company has decided to apply lean thinking, the transformation towards lean needs a comprehensive approach, and the result will be a lean enterprise. Lean transformation is a long-term activity which requires even years to be completed. An emergent strategy can be guided by an overall umbrella strategy, where the management sets out the vision and the broad guidelines for a strategy; detailed specifics can be created at lower levels in the organisation [10], [11]. A transformation needs both strategy-oriented and operations-oriented approaches [15].

MIT researchers [13] have created the "Transition-to-Lean Roadmap", which is aimed at assisting firms in their efforts to transform into lean. The framework includes the steps necessary to initiate, sustain, and continuously refine an enterprise transformation based upon lean principles and practices. This model was developed from an enterprise perspective and is illustrated in Fig. 1.

III. TOWARDS A NEW FRAMEWORK FOR LEAN TRANSFORMATION

Despite the fact that lean paradigm has been enhanced also to software industry the current literature does not provide proper lean transformation frameworks for software companies; the existing frameworks are intended mainly to manufacturing industries. Additionally, the turbulent business environment is a remarkable driver for software companies to transform, and the cloud adds challenges ever more. The capacity of an organisation to benefit from the opportunities created by external factors, e.g. the cloud, depends on its ability to adapt the entire organisation to the change. The missing lean transformation framework for software companies and the challenges created by the cloud led us to create an appropriate lean transformation framework for software intensive companies. Based on the frameworks presented ([7], [10], [12], and [13]) we created a revised framework that is shown in Fig. 2. The proposed lean transformation framework includes three cycles i.e. Strategic alignment cycle, Organisational and business alignment cycle, and Lean implementation cycle.

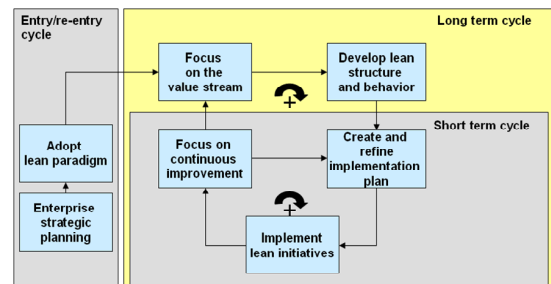


Figure 1. A framework for lean transformation (modified from [13]).

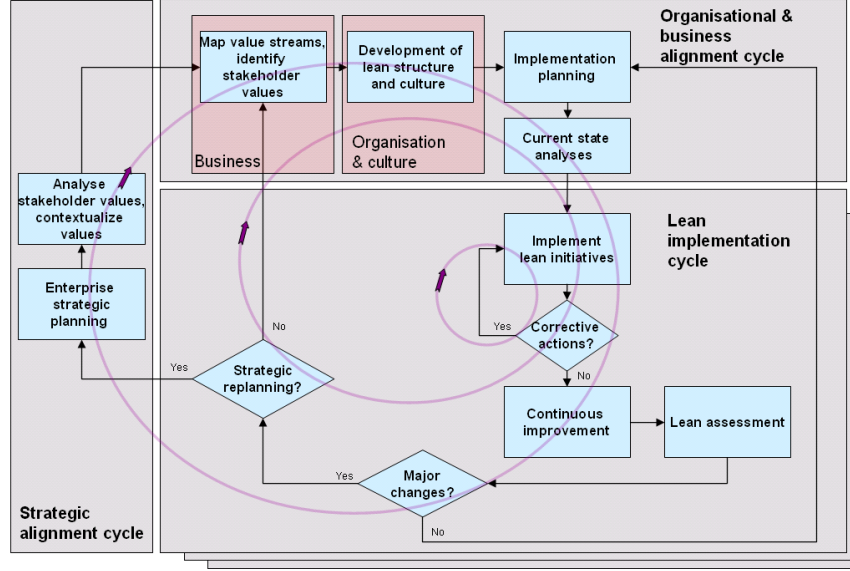


Figure 2. A new framework for lean transformation.

A. Strategic Alignment Cycle

The transformation cycle requiring the longest running time in the lean transformation is called Strategic alignment cycle, which corresponds to MIT [13] roadmap's Entry/Re-entry cycle (Fig. 1). This cycle concerns the highest organisational level of the enterprise, where strategic planning takes place. As lean paradigm focuses on the whole enterprise, lean thinking shall be intertwined to the strategic planning; not to be understood as a discrete element that can be added later on top of the plan. As well, lean thinking highlights the importance of customer values.

Stakeholders' values shall be contextualised in an early phase of transformation and the values shall yield through the whole transformation. Strategic alignment cycle considers external factors, i.e. drivers such as customer needs, business drivers, and especially the cloud. In MIT's [13] "Transition-to-Lean Roadmap", the researchers highlight the importance of adopting lean paradigm including tasks like build vision, convey urgency, foster lean learning, make the commitment, and obtain senior managers buy-in. All this is seen important also in our framework. However, our framework foregrounds also all stakeholders' values.

TABLE I. ACTIVITIES IN STRATEGIC ALIGNMENT CYCLE

Activity	Description
Strategic planning	An enterprise's strategy is a response to the challenges created by external factors (e.g. cloud) in order to gain or maintain the competitive advantage of the enterprise. Lean paradigm shall be intertwined in the strategy. Commitment of senior managers is a vital enabler of successful implementation of the strategy.
Contextualising values	The foundation of lean thinking is on creating value for stakeholders. The enterprise shall be able to understand and capture the values that will be pursued.

B. Organisational and Business Alignment Cycle

As soon as the main strategic goals of the organisation have been agreed, activities of Organisational and business alignment cycle has sufficient inputs to start. This cycle corresponds to MIT [13] roadmap's Long term cycle (Fig. 1). In this cycle, both business and organisational development related things shall be planned and deployed.

MIT researchers [13] highlight the importance of value stream by proposing value stream mapping, creating a vision of

TABLE II. ACTIVITIES IN ORGANISATIONAL AND BUSINESS ALIGNMENT CYCLE

Activity	Description
Focus on value stream	Lean thinking focuses on value creation for stakeholders. Mapping the value stream shall start from customers and move inwards to the enterprise processes. The enterprise's revenue models shall be studied and market forecasts shall be considered, and further, different business cases shall be analysed.
Cultural and organisational development	Lean transformation requires aligning the whole organisation to support it. The enterprise's processes and practices shall be aligned with lean principles. The enterprise culture should support lean philosophy by encouraging and making possible to create initiatives and innovations.
Planning lean implementation	Planning lean implementation needs a roadmap, including rough estimate of schedule and resources. Trainings at all levels of the company are vital. These trainings also help in changing the enterprise culture.
Current state analysis (lean)	Current state analyses help focus on the most urgent process areas or units of the organisation. These analyses are conducted using lean principles as the framework of the activity.

how customers pull value from the value stream, and by involving stakeholders in the transformation. We agree, and additionally, we propose further examining customer values, value networks, business case analyses and revenue models. This is extremely important in the current turbulent business environment.

Organisational structure shall support the business plan deployment, too. And still, in order to properly implement the lean transformation the enterprise shall consider cultural aspects. This is aligned with MIT's [13] roadmap. MIT researchers propose horizontal focus in organisational structure, empowering change agents and aligning incentives to support lean implementation.

In MIT's [13] roadmap (Fig. 1), creating lean implementation plan is located in the Short-term cycle. However, they say that lean implementation framework is a kind of transition element between Long-term and Short-term cycles. MIT [13] roadmap includes transitioning between roadmap cycles, and the researchers say that although the flow pattern suggests strict linearity, some of the activities may be performed in parallel. Thus, interpretation of the roadmap may have different views.

We propose creation of lean implementation plan at enterprise level belonging to Organisational and business alignment cycle, i.e. long-term cycle, as well as analysing the current state of selected processes or functions.

C. Lean Implementation Cycle

Lean implementation cycle can start, when main strategic, organisational and business alignment decisions have been made and current status of operational mode is defined. This cycle corresponds to MIT [13] roadmap's Short term cycle (Fig. 1). Lean implementation represents shorter term activities and focuses on continuous improvement at team or unit level. At intervals, lean assessments are useful to get a picture how lean transformation is progressing. From time to time it is vital to check if corrective actions are needed, major or minor ones. Even though strategic planning aims at longer term plans, it is useful to check if the enterprise's strategic plans should be revisited and updated to respond to so far unforeseeable changes.

In most (large) companies, there are own departments or units for different functions, e.g. research and development, human resources, accounting, marketing etc. This is illustrated

by several instances of Lean implementation cycle in Fig. 2. Lean implementation takes place side by side in several places of the organisation. Our Lean implementation cycle and MIT [13] roadmap's short term cycle are pretty well aligned with each other.

IV. APPLICABILITY OF THE NEW FRAMEWORK

In this chapter, applicability of the framework for a software company is discussed. The software company in question is a large service integrator company Sintco (pseudonym), who started a systematic Cloud transformation at the end of the first decade of 2000. Company has almost 20000 employees in 20+ countries, most of them in Northern and Central Europe, Russia, India and China. Most of the employees are located in Finland and sales come from Scandinavian countries.

Sintco's cloud transformation is all-embracing initiative that touches almost all business units in one way or another. Big part of Sintco's business is coming from the IT operations and application management and in there the impact and emergence of the new cloud-based services is understandably biggest.

At the same time with the enterprise-level cloud transformation Sintco started to actively map and analyse its existing operations, especially their "cloudability". One of the services that was seen to have very good potential to be full blown cloud service, was one of the Sintco's sustainability related service and its development; Susde (pseudonym). In the following chapters, applicability of the framework is discussed in the context of Susde's attempt to adopt lean way of working in their cloud-based service development and how it would match with the overall framework. This means, that the cloud transformation of Sintco is seen to be at the background, and the framework is mapped on top of that via operations of Susde.

With Susde, the iterative nature of the framework was all the time present in their transformation. It should be noted though that in Susde's case, first step (1. Enterprise strategic planning including lean thinking philosophy) did not include so much enterprise's *own* articulated lean thinking as such, but focus was more on "text book" approach for the applying lean and agile way of working.

Regarding the next steps of the framework, the Susde's development team followed rather meticulously, but again, with their own business interest being the basic principle and criteria for the selection of improvement actions. In practice that meant, that Susde's development team selected their requirement management process to be their first improvement area, and their lean related activities like value stream mapping, customer value identification, etc. were first targeted to that specific area of their business, since it was felt to bring biggest value for their business.

TABLE III. ACTIVITIES IN LEAN IMPLEMENTATION CYCLE

Activity	Description
Implementing lean initiatives	Based on the enterprise level implementation plan, more detailed plans shall be created and their implementation shall start.
Continuous improvement	In this short-term cycle focus is on continuous improvement i.e. Kaizen.
Lean assessment	In order to get information of the progress of realisation of the implementation initiatives, lean assessments will be conducted when appropriate.

V. DISCUSSION

This paper proposes the lean transformation framework for software intensive companies (Fig. 2) based on the lean principles and the related frameworks of the current literature. Our framework validates pretty well the “Transition-to-Lean Roadmap” of MIT [13], which is primarily intended to manufacturing companies, though.

The Poppendiecks [9] propose a roadmap for lean software development with ten steps. Table IV that links them to our framework indicates that all the steps of the roadmap [9] are covered in our framework. Additionally, our framework proposes to consider the organisational structure when transforming to lean. As well, the proposed framework suggests to study the whole value stream, not just customer values, and to consider revenue models and market forecast, and to conduct business case analyses. Our framework covers wider perspective than the Poppendiecks’ roadmap [9], which can be interpreted to cover only the organisation’s internal environment; we propose a wider view.

What comes to the *scope* of the transformation, the proposed framework considers at the end the whole enterprise. However, at the time when writing this paper, most practical and concrete initiatives have been taken in a more concise context, i.e. in research and development functions and their activities. This applies also to the *means* of the transformation. Still, this is understandable as the transformation in the case company is in its early phases of a long journey.

TABLE IV. MAPPING STEPS OF THE ROADMAP OF THE POPPENDIECKS [9] TO THE PROPOSED LEAN TRANSFORMATION FRAMEWORK

The Roadmap of the Poppendiecks [9]	The Proposed Lean Transformation Framework (Fig. 2)
1. As a starting point consider how to create value and make a profit.	Strategic alignment cycle; - Enterprise strategic planning including lean thinking philosophy
2. Think what is the biggest problem limiting ability to create value and make a profit.	Strategic alignment cycle; - Analyse stakeholder values, contextualise values
3. Find out what is the biggest threat to ability to create value and make a profit over the long-term.	Organisational and business alignment cycle; - Focus on value stream, identify customer values; - Current state analyses
4. Evaluate the culture.	Organisational and business alignment cycle; - Develop lean structure and culture
5. Train people.	Organisational and business alignment cycle; - Create and refine implementation plan
6. Solve the biggest problem, turn the biggest constraint over to work teams.	Organisational and business alignment cycle / Lean implementation cycle
7. Change rules in order to solve the problems and to remove constraints.	Lean implementation cycle; - Implement lean initiatives - Focus on continuous improvement
8. Measure to see the progress.	Lean implementation cycle; Lean assessment
9. Adopt changes supported by results.	Lean implementation cycle; Focus on continuous improvement
10. Repeat the cycle, start again from step 1.	Feedback loops of the framework

VI. CONCLUSIONS AND FURTHER WORK

This paper has outlined a lean transformation framework, which was created in the context of the cloud. However, one has to keep in mind that the cloud is only one possible driver for big changes in contemporary companies. The case company is just taking their early steps in the transformation; thus, this framework is very much a work in progress, as well the transformation will take several years until firm evidence is available. However, experiences so far show very promising.

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